Six Birds for Navier–Stokes: A Mechanized No-Zeno Scaffold

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Abstract

Energy estimates for the three-dimensional incompressible Navier–Stokes equations bound the kinetic energy, but they do not stop velocity gradients from concentrating in ever smaller regions. This is why the classical regularity criteria, such as Beale–Kato–Majda [2], ask for control of a peak quantity. This paper builds a multiscale *decision scaffold*, inspired by the Six Birds framework, around a different question: can activity cascade to arbitrarily fine scales in finite time, a *Zeno cascade*? The scaffold keeps three kinds of statement apart.

Proved in Lean. If every refinement step costs a fixed quantum of work, and the work available per unit time at depth j is bounded by a capacity whose reciprocals have divergent sum, then no Zeno cascade occurs. Route-stable reductions have at most polynomial capacity growth. In a finite model, a uniform capacity certificate over inputs that include localized tests forces a pointwise bound on the gain.

Proved on paper, or stated as open hypotheses. We define the subgrid bridge with sharp Fourier shells and show that on a strong solution it returns the actual closure force. Energy alone bounds the power that the bridge supplies to shell j by $C\,2^{5j/2}\|u_0\|_2^3$. That bound is too weak to certify divergence. An explicit three-mode field shows that no bound of the form $\|\nabla u\|_\infty \|u_j\|_2^2$ holds for this power.

Measured. An anti-localization certificate on initial data passes band-limited random fields ($\max_j \kappa_j \approx 2.1$) and rejects a localized shell wavepacket ($\kappa \approx 404$).

We do not prove the Clay regularity statement, and we do not show that the certificate persists under the flow. The Navier–Stokes inequalities that a proof along these lines would still need are listed explicitly.

Keywords: Navier–Stokes; multiscale scaffold; No-Zeno engine; route capacity; anti-localization; peak-channel regularity; Lean; Six Birds framework

1 Introduction

The Clay Millennium problem for the three-dimensional incompressible Navier–Stokes equations asks whether smooth initial data always produce solutions that stay smooth for all time, or whether a finite-time singularity can form [5]. Energy methods alone have not settled the question. The kinetic energy stays bounded, but a bounded energy is compatible with gradients that become very large in very small regions. Conditional criteria make this precise: if a peak quantity, such as the maximum vorticity, is integrable in time, the solution cannot blow up [2, 10]. What is missing is a proof that such peak control holds for every smooth initial datum.

This paper does *not* solve the Clay problem. It builds a *decision scaffold*: a precise chain of statements, each labelled as proved, measured or open. The chain organizes one multiscale strategy

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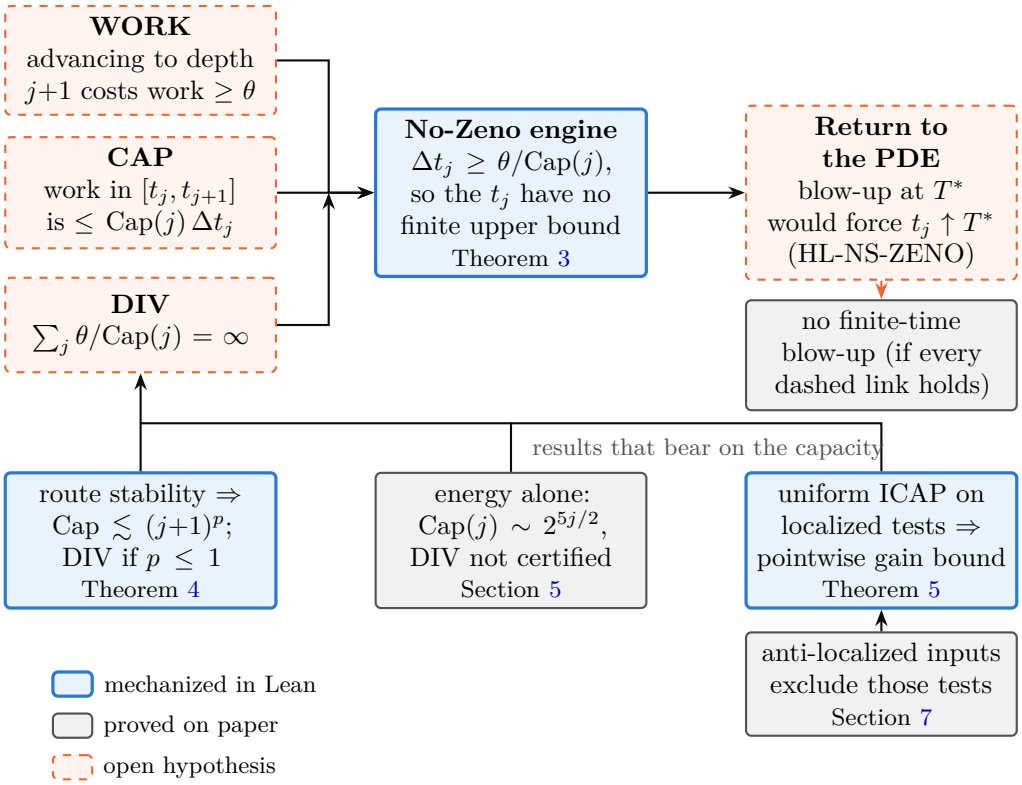


Figure 1: **Map of the scaffold.** Top row: the No-Zeno engine turns WORK, CAP and DIV into a statement about crossing times. A separate, open return lemma would turn that statement into regularity. Bottom row: the results that bear on the capacity. Route stability limits capacity growth to a polynomial, which certifies DIV (with a constant work quantum) when its degree is at most one. Energy alone gives a valid capacity that grows too fast to certify DIV. A uniform capacity certificate over localization-rich inputs is, in the finite model, as strong as a pointwise gain bound. Anti-localization removes the localized tests that drive that equivalence, but it does not supply a capacity bound by itself.

suggested by the Six Birds framework [14]. The idea is to treat the elimination of fine scales as a *packaging* operation. The eliminated degrees of freedom become a causal "bridge" that acts on the retained scales through an interface, and an energy ledger constrains what the bridge can do. In this picture a singularity would be a *Zeno cascade*: an activity front that descends through infinitely many dyadic scales in finite time.

The strategy in one paragraph. Label the dyadic scales $j = 0, 1, 2, \ldots$ and let t_j be the time at which a suitably defined frontier first reaches depth j. Suppose each advance costs at least a fixed amount of work $\theta > 0$ (WORK). Suppose also that the bridge can deliver work to depth j at a rate of at most Cap(j) (CAP). Then the step from depth j to depth $j+1$ takes at least $\theta/\mathrm{Cap}(j)$ units of time. If these lower bounds have divergent sum (DIV), the crossing times cannot accumulate at a finite time. The logic is elementary and is mechanized here. The difficulty lies in producing WORK, CAP and DIV for the actual Navier–Stokes bridge, and in showing that a blow-up would really push this frontier to infinite depth. Figure 1 shows the resulting chain and where each link stands.

What the analysis shows. Two findings shape the rest of the paper. First, a capacity certificate that holds uniformly over all admissible inputs is an extremal bound. In a finite model, if the admissible inputs include localized test vectors, such a certificate is exactly as strong as a pointwise bound on the gain (theorem 5). That is the shape of a classical peak criterion. We do not prove the continuum analogue for the actual bridge (section 6 lists what it would need), but the finite theorem marks the risk: unless the admissible inputs are restricted, a uniform capacity certificate may be no weaker than peak control. Second, the information that is always available is too weak to close the chain. Energy alone gives a capacity that grows like $2^{5j/2}$ in the depth j, and the reciprocals of such a capacity have a finite sum. The natural sharpening, a bound by $\|\nabla u\|_\infty$ times the shell energy, fails for the full subgrid port: an exact three-mode field gives supplied power proportional to ε with shell energy proportional to ε^2 (proposition 2).

A restricted problem. These findings suggest one escape route inside the scaffold, which we adopt as a proposal: require admissible inputs to be spread out in space at each scale, so that localized tests are not available. We call such data *SBT-legal* ("SBT" for Six Birds Theory). We state the corresponding *anti-localization* hypothesis and prove, in Lean, that it follows from a finite set of spread-out channels. We also turn it into a reproducible certificate for initial data. Smooth random band-limited fields pass the certificate. A band-limited wavepacket concentrated in one small ball exceeds the threshold forty-fold, about two hundred times the baseline's value. Whether the Navier–Stokes flow preserves legality is open.

Three layers of claims. Every statement in the paper belongs to one of three layers.

- **Mechanized (Lean).** The inequality logic of the No-Zeno engine, the route-capacity bound, the capacity aggregation step, the finite anti-localization lemma and the finite no-go theorem. These are system-agnostic statements about sequences and finite sums. The 31 audited declarations depend only on the standard axioms `propext`, `Classical.choice` and `Quot.sound` (appendix A).
- **Analytic (paper proofs).** Navier–Stokes-facing statements with complete proofs in the text: the exact return of the bridge, the energy bound on the subgrid power and its counterexample, the residual-charged anti-localization bound, and the properties of the regularized model used in the simulations (appendix B).
- **Open or measured.** The hinge lemmas that a regularity proof along these lines would need are stated as open hypotheses. Numerical diagnostics are reproducible from a manifest. They are evidence, not proofs.

Contributions.

- A mechanized No-Zeno chain and a mechanized route-capacity theorem, with the Lean adapters that connect each theorem to the form in which it is used (sections 3 and 4).
- A precise packaged-bridge definition for Navier–Stokes, with the sign of the supplied power fixed and an exact return on strong solutions (section 2).
- The energy-only capacity bound and an explicit counterexample to a shell-only strain bound for the full subgrid port (section 5).

- A finite no-go theorem that shows why uniform capacity certificates and localization-rich inputs do not mix, with the continuum passage left explicitly open (section 6).
- An anti-localization hypothesis, its finite-channel proof, and an operational certificate with a manifest-driven, deterministic figure pipeline (sections 7 and 8).

Reading guide. Section 2 recalls the Six Birds vocabulary and fixes the Navier–Stokes objects. Sections 3 and 4 give the mechanized engine and the route-capacity theorem. Section 5 shows what energy alone provides. Section 6 states the no-go theorem. Section 7 develops anti-localization and the certificate, and section 8 reports the diagnostics. Section 9 lists what remains open. The appendices contain the Lean index, the regularized model, the numerical method, the reproduction commands and a version note.

2 Six Birds vocabulary and the Navier–Stokes objects

This section introduces the parts of the Six Birds framework [14] that the scaffold uses, and it fixes the Navier–Stokes objects that later sections refer to. For standard Navier–Stokes background see Temam [12] or Majda and Bertozzi [7].

2.1 Six moves for multiscale elimination

The Six Birds framework describes six recurring structural moves. Here they serve as a vocabulary for eliminating fine scales without committing to a closure model.

P1, rewrite. Choose coordinates in which scale is explicit. We use dyadic Fourier shells. This is a change of description, not a modelling assumption.

P2, feasibility. Decide which interface signals are admissible. Feasibility appears twice: in the domain of the packaged bridge, and in the anti-localization restriction of section 7.

P3, route dependence. Eliminating scales in different orders can give different effective descriptions. Section 4 turns a bound on this mismatch into a bound on capacity growth.

P4, sectorization. Interactions may decompose into a small number of channels. We treat the channel count as a measurable quantity and prove that few spread-out channels force anti-localization.

P5, packaging. Turn everything above a cutoff into one causal object, the bridge, that acts on the retained scales. The bridge is defined below; it is not fitted.

P6, accounting. Constrain the bridge by an energy ledger, in the spirit of dissipativity theory [15]. The currency is the positive work that the bridge supplies to a shell.

2.2 Shells, projections and the closure force

Work on the periodic box $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with zero-mean, divergence-free velocity fields, viscosity $\nu > 0$ and no external force. For $j \geq 0$ let

$$\mathcal{K}_j = \{k \in \mathbb{Z}^3 : \ 2^j \leq |k| < 2^{j+1}\}.$$

Let Δ_j be the *sharp* orthogonal Fourier projection onto the modes in $\mathcal{K}_j$, let G_j project onto $|k| < 2^{j+1}$, and set $H_j = I - G_j$. Every nonzero frequency on this box has $|k| \geq 1$, so a zero-mean field decomposes as $u = \sum_{j\geq 0} \Delta_j u$. We write

$$u_j := \Delta_j u, \qquad u_{\leq j} := G_j u, \qquad u_{>j} := H_j u.$$

Because the projections are sharp, $\Delta_j^2 = \Delta_j$, $G_j^2 = G_j$, $\Delta_j G_j = \Delta_j$ and $G_j H_j = 0$, and all of them commute with derivatives and with the Leray projection $\mathcal{P}$. Smooth, overlapping Littlewood–Paley multipliers [1] do not have these identities. A smooth-filter version of the constructions below would need a separate argument.

Write $\mathrm{NL}(u) := \mathcal{P}\nabla \cdot (u \otimes u)$, so that a strong solution satisfies $\partial_t u = \nu \Delta u - \mathrm{NL}(u)$. The *closure force* at cutoff j and its shell readout are

$$C_{\leq j}(u) := G_j \,\mathrm{NL}(G_j u) - G_j \,\mathrm{NL}(u), \qquad y_j(u) := \Delta_j C_{\leq j}(u). \tag{1}$$

With these signs the exact retained and shell equations read

$$\partial_t u_{\leq j} = \nu \Delta u_{\leq j} - G_j \mathrm{NL}(u_{\leq j}) + C_{\leq j}(u), \qquad \partial_t u_j = \nu \Delta u_j - \Delta_j \mathrm{NL}(u_{\leq j}) + y_j(u).$$

Thus y_j is the force that the eliminated scales add to shell j, and

$$p_j(t) := \langle u_j(t), y_j(t) \rangle_{L^2}, \qquad p_j^+ := \max(p_j, 0), \qquad W_j^+[s,t] := \int_s^t p_j^+(\tau)\, d\tau \tag{2}$$

are the power *supplied* to the shell, its positive part, and the positive supplied work. The sign matters. Reversing it exchanges supplied and extracted power and changes W_j^+.

2.3 The packaged bridge

The bridge packages the eliminated scales as a causal map on resolved histories.

Definition 1 (Packaged bridge Z_j)**.** Fix $s > 5/2$, $T > 0$ and $u_0 \in H^s$. A resolved history $v \in C([0,T]; \operatorname{Ran} G_j)$ is *admissible* if the high-mode problem

$$\partial_t w = \nu \Delta w - H_j \,\mathrm{NL}(v + w), \qquad w(0) = H_j u_0, \tag{3}$$

has a unique strong solution $w \in C([0,T]; H^s) \cap L^2(0,T; H^{s+1})$. For admissible v set

$$Z_j[v](t) := \Delta_j \big(\mathrm{NL}(v(t)) - \mathrm{NL}(v(t) + w(t))\big).$$

The input of Z_j is the whole resolved history v, not only its shell-j part. The power pairs the output with $\Delta_j v$. Changing other resolved shells can change the output while leaving $\Delta_j v$ untouched, so the bridge does not in general factor through a single-port map. We do not claim that every candidate history is admissible.

Proposition 1 (Causality and exact return)**.**

1. *The high-mode range is invariant: a solution of (3) satisfies $H_j w = w$.*
2. *If two admissible histories agree on $[0,t]$, the corresponding high-mode solutions and bridge outputs agree on $[0,t]$.*
3. *If u is a strong Navier–Stokes solution on $[0,T]$, then $v = G_j u$ is admissible and $Z_j[G_j u](t) = y_j(u(t))$ for $t \in [0,T]$.*

Proof. (1) Applying G_j to (3) leaves a heat equation for $G_j w$ with zero data, so $G_j w = 0$. (2) Strong solutions on $[0,t]$ are unique. For two solutions w_1, w_2 driven by the same history, the difference z obeys an L^2 energy estimate. The transport term with the full velocity cancels, H_j drops out of the pairing with z, and the remaining terms are bounded by the strong gradients of v, w_1, w_2; Grönwall gives $z = 0$. (3) Applying H_j to the Navier–Stokes equation shows that $w = H_j u$ solves (3) with $v = G_j u$, and uniqueness identifies it with the packaged high state. Since $\Delta_j = \Delta_j G_j$, the definition of Z_j then coincides with $y_j(u)$ in (1). □

Frequency-response fits of the subgrid term are finite-window, linear time-invariant approximations. They neither define Z_j nor certify any property of it over its admissible class.

2.4 Capacity and frontier

Two further objects connect the bridge to the No-Zeno engine of section 3.

Capacity. The integrated capacity inequality (ICAP) at depth j is

$$W_j^+[s,t] \;\le\; \Lambda(j) \int_s^t \|u_j(\tau)\|_{L^2}^2 \, d\tau \qquad \text{for all admissible inputs and windows } [s,t]. \tag{4}$$

It is a certificate quantified over the admissible class, not a bound along one trajectory. If in addition $\|u_j(t)\|_{L^2}^2 \le B(j)$, then $W_j^+[s,t] \le \mathrm{Cap}(j)\,(t-s)$ with $\mathrm{Cap}(j) = \Lambda(j)B(j)$. For unforced Navier–Stokes the energy inequality gives $B(j) = \|u_0\|_{L^2}^2$ at every depth.

Frontier. Let $S_j \ge 0$ be a storage functional for the eliminated scales, continuous in time, and assume the passivity inequality $S_j(t) - S_j(s) \le \int_s^t p_j \le W_j^+[s,t]$. Given a threshold $\theta > 0$, the frontier crosses from depth j to depth $j+1$ at

$$t_{j+1} := \inf\Big\{ t \ge t_j : \sup_{\tau \in [t_j, t]} \big(S_j(\tau) - S_j(t_j)\big) \ge \theta \Big\}.$$

Writing an infimum does not make the crossing exist. We call a crossing *attained* when the set is nonempty and t_{j+1} is finite and lies in the time domain. A frontier index $j_*(t) = \sup\{j : t_j \le t\}$ may be infinite; assuming that it is a finite maximum would assume away the Zeno accumulation that the scaffold is meant to rule out.

The storage functional, the passivity inequality and the capacity bound are hypotheses about the actual bridge. None of them is proved for Navier–Stokes here.

3 The No-Zeno engine

This section isolates the inequality logic of the scaffold. Nothing in it is specific to Navier–Stokes. It applies to any multiscale process whose refinement events come with an increment lower bound and a divergence check. The theorems are mechanized in Lean in a discrete setting: crossing times form a sequence $t : \mathbb{N} \to \mathbb{R}$ and all hypotheses are stated with finite sums. The continuous-time arguments that would produce those hypotheses are separate. The one used here is proved on paper in lemma 1.

3.1 Crossing times and Zeno cascades

Let $t : \mathbb{N} \to \mathbb{R}$ be a sequence of crossing times, where $t(n)$ is read as the time at which the frontier reaches level n, and write $\Delta t_n := t(n+1) - t(n)$.

Definition 2 (Zeno cascade)**.** A *Zeno cascade* occurs if the crossing times are bounded above: there is $T \in \mathbb{R}$ with $t(n) \le T$ for every n. In words, the frontier reaches every level before a finite time.

Definition 3 (Increment certificate)**.** A sequence $\Phi : \mathbb{N} \to \mathbb{R}$ is an *increment certificate* for t if $\Phi(n) \le t(n+1) - t(n)$ for every n.

Theorem 1 (Unbounded partial sums exclude a Zeno cascade)**.** *Let Φ be an increment certificate for t whose partial sums are unbounded: for every $M \in \mathbb{R}$ there is n with $M \le \sum_{k=0}^{n-1} \Phi(k)$. Then t has no finite upper bound, so no Zeno cascade occurs.*

Proof. The increment inequality telescopes to $t(n) \ge t(0) + \sum_{k=0}^{n-1} \Phi(k)$, and the right-hand side is unbounded. □

If, in addition, every $\Phi(n)$ is nonnegative, then t is nondecreasing; Lean proves this as a separate lemma. The theorem itself needs neither monotonicity nor nonnegativity.

3.2 WORK, CAP and DIV

In applications the increment certificate comes from a work quantum and a capacity.

Definition 4 (WORK/CAP/DIV)**.** Let $w, \mathrm{Cap} : \mathbb{N} \to \mathbb{R}$ with $\mathrm{Cap}(n) > 0$ for every n.

- **WORK×CAP:** $w(n) \le \mathrm{Cap}(n)\,(t(n+1) - t(n))$ for every n. Here $w(n)$ is the work needed to advance from level n to level $n+1$, and $\mathrm{Cap}(n)$ bounds the work that can be delivered per unit time at level n.
- **DIV:** the ratios $w(n)/\mathrm{Cap}(n)$ have unbounded partial sums.

Theorem 2 (Work and capacity give an increment bound)**.** *If $\mathrm{Cap}(n) > 0$ and $w(n) \le \mathrm{Cap}(n)\,(t(n+1) - t(n))$, then $w(n)/\mathrm{Cap}(n) \le t(n+1) - t(n)$.*

Theorem 3 (WORK, CAP and DIV exclude a Zeno cascade)**.** *Under WORK×CAP and DIV, the crossing times have no finite upper bound.*

Proof. Theorem 2 makes w/Cap an increment certificate, and theorem 1 applies. □

Dyadic sampling. Sometimes crossings are only checked at each doubling of scale. Lean also proves the sampled version: if $\mathrm{Cap}(2^j) > 0$, $w(2^j) \le \mathrm{Cap}(2^j)\,(t(j+1) - t(j))$ for every j, and the ratios $w(2^j)/\mathrm{Cap}(2^j)$ have unbounded partial sums, then $j \mapsto t(j)$ has no finite upper bound.

3.3 Where the hypotheses would come from

The engine is only as useful as its inputs. Two steps connect it to the bridge of section 2.

Lemma 1 (A storage frontier forces a work quantum)**.** *Assume the storage S_j is continuous, the power p_j is locally integrable, the passivity inequality $S_j(t) - S_j(s) \le W_j^+[s,t]$ holds, and the crossing t_{j+1} defined in section 2.4 is attained. Then $W_j^+[t_j, t_{j+1}] \ge \theta$.*

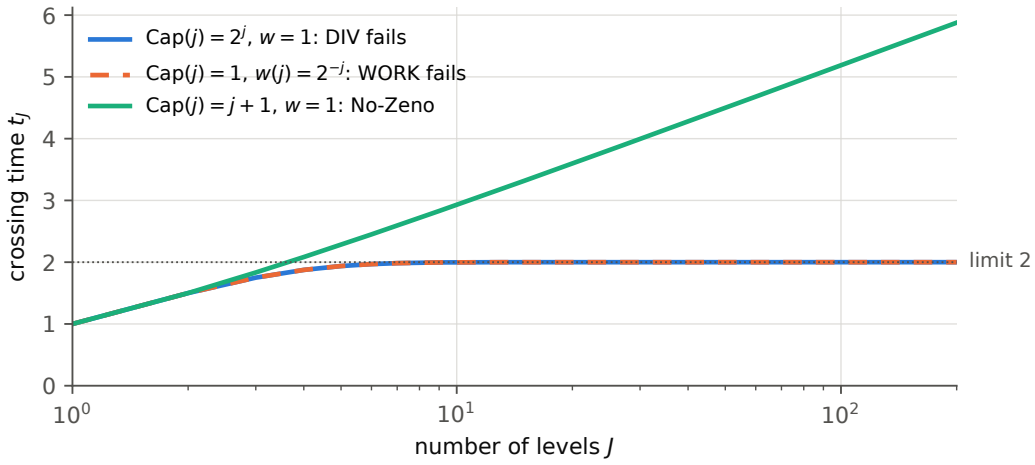


Figure 2: **Toy Zeno gallery.** Crossing times $t_J = \sum_{j<J} \Delta t_j$ for three explicit sequences with $\Delta t_j = w(j)/\mathrm{Cap}(j)$. With exponentially growing capacity (DIV fails) or a vanishing work quantum (WORK fails), the times converge to the same limit 2 (both have $\Delta t_j = 2^{-j}$, so their curves coincide) and a Zeno cascade occurs. With linear capacity and constant work, DIV holds and t_J grows without bound. The figure is purely mathematical; it illustrates why both WORK and DIV are needed.

Proof. The running maximum $A(t) = \sup_{\tau \in [t_j, t]}(S_j(\tau) - S_j(t_j))$ is continuous, so $A(t_{j+1}) \geq \theta$. By compactness some $\tau^* \in [t_j, t_{j+1}]$ attains it, and $\theta \leq S_j(\tau^*) - S_j(t_j) \leq W_j^+[t_j, \tau^*] \leq W_j^+[t_j, t_{j+1}]$, since $p_j^+ \geq 0$. □

Combining lemma 1 with a capacity bound $W_j^+[t_j, t_{j+1}] \leq \mathrm{Cap}(j)\,\Delta t_j$ gives WORK×CAP with $w(j) = \theta$. The lemma is a continuous-time argument. It is not part of the Lean development, which only performs the division and summation steps.

Checking DIV. DIV is a statement about a series, so only the right comparisons decide it. With a constant work quantum $\theta > 0$:

- if $\mathrm{Cap}(j) \leq K(j+1)^\alpha$ eventually, with $K > 0$ and $\alpha \leq 1$, then $\sum_j \theta/\mathrm{Cap}(j) = \infty$ and DIV holds;
- if $\mathrm{Cap}(j) \geq k(j+1)^\alpha$ eventually with $k > 0$ and $\alpha > 1$, or $\mathrm{Cap}(j) \geq k\,2^{\beta j}$ with $k, \beta > 0$, then the series converges and this capacity cannot certify DIV.

An *upper* bound that grows faster than linearly decides nothing: $\mathrm{Cap}(j) = 1$ satisfies every such bound and has a divergent reciprocal series. Conversely, a convergent comparison series only leaves No-Zeno unproved. It does not show that the actual crossing increments have a finite sum, and it does not show that blow-up occurs.

Remark 1 (Mechanization)**.** Theorem 1 and the monotonicity lemma are in `formal/Fluids/Zeno.lean`; theorems 2 and 3 and the dyadic version are in `formal/Fluids/ZenoWorkCap.lean`. Lean names are listed in appendix A. The development uses Lean 4 and mathlib [13, 4, 8].

4 Route stability limits capacity growth

DIV needs a capacity that grows slowly with depth. One place where the Six Birds viewpoint suggests leverage beyond one-shot estimates is *route dependence*. A route is an order in which scales are eliminated. If eliminating two scales at once is close to eliminating them one at a time, the capacity cannot inflate quickly from one depth to the next. This section turns that idea into an abstract inequality and proves it in Lean.

4.1 Capacity recursion with mismatch

Let $C : \mathbb{N} \to \mathbb{R}$ be a *nonnegative* sequence, read as the effective capacity after n reduction steps, and let $e : \mathbb{N} \to \mathbb{R}$ be a nonnegative mismatch sequence in the same currency. Fix $p \in \mathbb{N}$ and assume the one-step recurrence

$$C(n+1) \;\le\; r(n)\, C(n) + e(n), \tag{5}$$

where the step factor is *tame*:

$$r(n) \;\le\; \Big(\frac{n+2}{n+1}\Big)^{p}. \tag{6}$$

The tame factors telescope, $\prod_{k=0}^{n-1}((k+2)/(k+1))^{p} = (n+1)^{p}$, so multiplicative inflation along the ladder is at most polynomial. The remaining question is whether the additive mismatch accumulates. We ask for bounded partial sums:

$$\sum_{k=0}^{n-1} e(k) \;\le\; E \qquad \text{for all } n, \text{ for some } E \ge 0. \tag{7}$$

Theorem 4 (Route stability gives polynomial capacity)**.** *Let $C \ge 0$ and $e \ge 0$ satisfy (5) with tame factors (6). Then for all n*

$$C(n) \;\le\; \Big(C(0) + \sum_{k=0}^{n-1} e(k)\Big)(n+1)^{p}.$$

If moreover (7) holds, then $C(n) \le (C(0)+E)(n+1)^{p}$, so C grows at most polynomially in the depth.

Proof. Because $C(n) \ge 0$, the tame bound may replace $r(n)$ in (5): $C(n+1) \le ((n+2)/(n+1))^{p} C(n) + e(n)$. Iterating, each mismatch term is transported forward by later tame factors, whose products are at most $(n+1)^{p}$, and the bound follows by induction. □

Nonnegativity of every $C(n)$ is used in the first step. Without it, an upper bound on $r(n)$ does not bound $r(n)C(n)$.

With $p \le 1$ and a constant work quantum, theorem 4 produces a capacity that satisfies the DIV comparison of section 3. For $p \ge 2$ the bound alone does not decide DIV.

Remark 2 (Mechanization)**.** Theorem 4 is in `formal/Fluids/RouteCapacity.lean`. The lemmas `route_capacity_poly_bound` and `route_capacity_poly_bound_of_bounded_sum` assume the recurrence with the tame factor already substituted. The adapters `route_capacity_poly_bound_of_tame_factor` and `route_capacity_poly_bound_of_tame_factor_of_bounded_sum` start from an actual factor $r(n)$ and use $C(n) \ge 0$ to make the substitution.

4.2 Aggregating channels

A capacity can also be assembled from parts. If the port at depth j splits into m_j atoms, with powers $a_1, \dots, a_{m_j}$ summing to the total power, then pointwise in time $\max(\sum_r a_r, 0) \le \sum_r \max(a_r, 0)$. If every atom satisfies ICAP with the same constant Λ_0, the aggregate satisfies ICAP with $\Lambda(j) \le m_j \Lambda_0$. Both steps are mechanized (`max_sum_le_sum_max`, `icap_sum_le_card_mul` in `formal/Fluids/ECTCap.lean`). For unforced Navier–Stokes, $B(j) = \|u_0\|_{L^2}^2$ then gives $\mathrm{Cap}(j) \le \|u_0\|_{L^2}^2 \Lambda_0 m_j$. With $m_j \le C_0(j+1)$ this is linear in j, and DIV holds. No Navier–Stokes atom decomposition with a uniform per-atom constant is constructed here; that is an open hypothesis.

4.3 Interpretation

In Six Birds language, (5) says that each additional depth step may inflate the capacity by a tame factor and pays a route-mismatch penalty for the noncommutativity of reductions. Condition (7) says that this mismatch does not build up, so there is no hidden "holonomy" that grows with depth. Theorem 4 proves no statement about the PDE. It shows that *if* a system admits route-stable packaging in this sense, exponential capacity growth is ruled out by algebra alone.

The repository includes diagnostics that estimate capacity-like sequences along a Navier–Stokes truncation ladder (`nsbu21_route_capacity_numbers.py`) and compare their step factors and mismatch with the hypotheses above (`nsbu23_route_capacity_hypothesis_check.py`). They use the supplied-power sign of (2) and an explicit two-stage nested composition. Their results are sampled, finite-window ratios. A decaying normalized mismatch, or a step factor within a fixed tolerance over finitely many depths, establishes neither (7) nor a polynomial law. Section 8 reports them with that status.

5 What energy alone gives the capacity

Before asking for new inequalities, it is worth knowing what the always-available information provides. For unforced Navier–Stokes this is the energy bound $\|u(t)\|_{L^2} \le \|u_0\|_{L^2}$. This section shows that energy gives a valid capacity, that this capacity grows too fast to certify DIV, and that a tempting sharper bound fails for the full subgrid port.

5.1 The energy bound

By (1), with $v = G_j u$ and $w = H_j u$,

$$y_j = \Delta_j \mathcal{P} \nabla \cdot F, \qquad F = v \otimes v - u \otimes u = -(v \otimes w + w \otimes v + w \otimes w).$$

The output is a finite Fourier sum over the $O(2^{3j})$ frequencies of shell j, each carrying one derivative of size $O(2^j)$, and each Fourier coefficient of F is bounded by $\|F\|_{L^1}$ up to the normalization of the box. Cauchy–Schwarz therefore gives

$$\|\Delta_j \mathcal{P} \nabla \cdot F\|_{L^2} \;\le\; C\, 2^{5j/2}\, \|F\|_{L^1}. \tag{8}$$

This uses only the finite output shell; no L^1 bound for the Leray projection is involved. Since $\|F\|_{L^1} \le 2\|v\|_{L^2}\|w\|_{L^2} + \|w\|_{L^2}^2 \le 3\|u\|_{L^2}^2$,

$$\|y_j\|_{L^2} \le C'\, 2^{5j/2} \|u\|_{L^2}^2, \qquad |p_j| \le C'\, 2^{5j/2} \|u\|_{L^2}^2\, \|u_j\|_{L^2} \le C'\, 2^{5j/2} \|u_0\|_{L^2}^3. \tag{9}$$

The last bound is an absolute bound on power, so for $u_0 \neq 0$ the choice $\mathrm{Cap}(j) = C' 2^{5j/2} \|u_0\|_{L^2}^3$ is a valid, positive capacity in the sense of WORK×CAP. (Zero data give the zero solution, which

needs no capacity.) It is not an ICAP coefficient. Dividing the middle bound by $\|u_j\|_{L^2}^2$ leaves a factor $1/\|u_j\|_{L^2}$, which can be arbitrarily large.

Consequence for DIV. For nonzero data and a constant work quantum, the comparison series $\sum_j \theta/(C'2^{5j/2}\|u_0\|_{L^2}^3)$ converges, so this capacity cannot certify DIV. By the discussion in section 3, this proves neither that the minimal capacity has a convergent reciprocal series nor that a sharper capacity is impossible. It shows only that energy alone does not close the chain.

5.2 A shell-only bound fails for the full port

A natural sharpening would bound the subgrid output by the strain and the shell itself:

$$\|y_j\|_{L^2} \lesssim \|\nabla u\|_{L^\infty}\|u_j\|_{L^2}, \qquad \text{hence} \qquad |p_j| \lesssim \|\nabla u\|_{L^\infty}\|u_j\|_{L^2}^2. \tag{10}$$

Commutator estimates do give bounds of this type for selected high–low interactions. They cannot cover the high–high part of F, which can feed a shell that is almost empty.

Proposition 2 (Exact counterexample)**.** *On $(\mathbb{R}/2\pi\mathbb{Z})^3$ let $k = (4,0,0)$, $l = (-3,1,0)$, $q = k + l = (1,1,0)$, $a = (0,1,0)$, $b = (1,3,0)$, $c = (1,-1,0)$, and for $\varepsilon > 0$*

$$u_\varepsilon(x) = a\cos(k\cdot x) + b\cos(l\cdot x) - \varepsilon\, c\sin(q\cdot x).$$

Then u_ε is smooth, zero-mean and divergence-free, $\|\nabla u_\varepsilon\|_{L^\infty}$ is bounded uniformly for $0 < \varepsilon \le 1$, and at $j = 0$

$$y_0(u_\varepsilon) = -\tfrac{3}{2}\, c\sin(q\cdot x), \qquad p_0 = \tfrac{3}{2}\,\varepsilon\,|\mathbb{T}^3|, \qquad \|u_{\varepsilon,0}\|_{L^2}^2 = \varepsilon^2|\mathbb{T}^3|.$$

Hence $p_0/\|u_{\varepsilon,0}\|_{L^2}^2 = 3/(2\varepsilon)$ is unbounded, and neither inequality in (10) holds with a constant independent of the field.

Proof. The three modes are divergence-free because $k\cdot a = l\cdot b = q\cdot c = 0$. On this box G_0 keeps the modes with $|k| < 2$, so $G_0 u_\varepsilon = u_{\varepsilon,0} = -\varepsilon c\sin(q\cdot x)$ and the modes k, l are eliminated. The retained field has no self-advection, and every interaction of the retained mode with an eliminated mode lands outside $\pm q$. The only contribution to shell 0 comes from the product of the two eliminated modes. The q-component of $(h\cdot\nabla)h$, with $h = a\cos(k\cdot x) + b\cos(l\cdot x)$, equals $-\frac{1}{2}\big((a\cdot l)\,b + (b\cdot k)\,a\big)\sin(q\cdot x) = -\frac{1}{2}(1,7,0)\sin(q\cdot x)$. Removing its component along q leaves $\frac{3}{2}c\sin(q\cdot x)$, so $\Delta_0\mathrm{NL}(u_\varepsilon) = \frac{3}{2}c\sin(q\cdot x)$ while $\Delta_0\mathrm{NL}(u_{\varepsilon,0}) = 0$. By (1) this gives $y_0 = -\frac{3}{2}c\sin(q\cdot x)$; pairing with $u_{\varepsilon,0}$ and using $|c|^2 = 2$ gives p_0 and the shell energy. □

Each u_ε is a smooth initial datum with a local smooth Navier–Stokes solution, so the counterexample lives in the instantaneous domain of the actual port. It refutes one specific estimate. It is not a no-go result for cumulative bounds over time windows, for the nonlinear packaged history class, or for other functionals. The finite no-go theorem of section 6 concerns a different, discrete statement and is unaffected.

6 Uniform capacity and localized tests do not mix

ICAP (4) is an extremal statement: it bounds the positive work over *every* admissible input. This section shows, in a finite model, what that costs. If the admissible inputs include localized test vectors, a uniform capacity certificate is exactly as strong as a pointwise bound on the gain. We then state what a continuum version for the actual bridge would require.

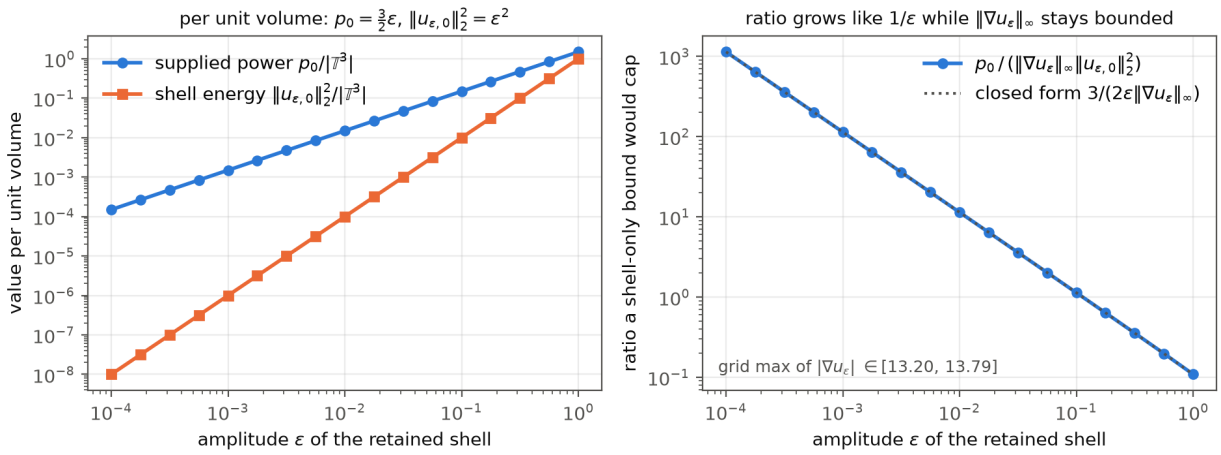


Figure 3: **The shell-only bound fails.** The field of proposition 2, evaluated with the repository's de-aliased pseudo-spectral nonlinearity on a 24^3 grid (integrals normalized by volume). Left: the supplied power falls like ε while the shell energy falls like ε^2. Right: the ratio that (10) would bound grows like $1/\varepsilon$, while the grid maximum of $|\nabla u_\varepsilon|$ stays between 13.2 and 13.8. The computed output matches $-\frac{3}{2}c\sin(q \cdot x)$ to 4×10^{-15}.

6.1 A finite obstruction

Let ι be a finite index set, such as the cells of a grid, and let $g : \iota \to \mathbb{R}$ be a gain profile with quadratic form $Q_g(u) = \sum_{i\in\iota} g(i)\, u(i)^2$. Let $\mathcal{F}$ be a class of feasible inputs $u : \iota \to \mathbb{R}$. A memoryless ICAP on $\mathcal{F}$ reads

$$\max(Q_g(u), 0) \;\le\; \Lambda \sum_{i\in\iota} u(i)^2 \qquad \text{for all } u \in \mathcal{F}. \tag{11}$$

Theorem 5 (ICAP on localized tests gives a pointwise gain bound)**.** *Assume* (11) *for some* $\Lambda \in \mathbb{R}$*, and assume* $\mathcal{F}$ *contains every coordinate vector* $\mathbf{1}_{\{i\}}$*. Then* $\max(g(i), 0) \le \Lambda$ *for every* $i \in \iota$*.*

Proof. Test (11) with $u = \mathbf{1}_{\{i\}}$, for which $Q_g(u) = g(i)$ and $\sum u^2 = 1$. □

The converse holds as well: if $\max(g(i), 0) \le \Lambda$ for every i and $\Lambda \ge 0$, then (11) holds for all u. On a localization-rich class, a uniform capacity certificate and a pointwise gain bound are therefore the same statement.

Mechanization. Lean proves the version with all inputs feasible (`icap_implies_pointwise_pos_bound_of_basis_tests`) and the version stated here, where ICAP is assumed only on a class that contains the basis tests (`icap_implies_pointwise_pos_bound_on_feasible_basis_tests`), both in `formal/Fluids/IcapNoGo.lean`.

6.2 What a continuum version would need

In the continuum the role of the basis vectors would be played by divergence-free wavepackets in shell j whose energy concentrates in a ball of radius about 2^{-j}. If the bridge acted on such packets through a local gain, a uniform ICAP would bound that gain pointwise in space. For Navier–Stokes the natural candidate is a strain-type gain, and a pointwise bound on it has the form of a classical peak criterion [7, 2, 10].

We do *not* prove this passage. The actual bridge is nonlinear and history-dependent, and the finite theorem cannot simply be applied to it. A continuum version would need three adapters, each open here:

1. an actual gain representation of the bridge's positive work on shell-j inputs;
2. feasibility of solenoidal localized test inputs in the bridge's admissible class;
3. a limit argument with control of the fixed background, turning localized averages at finite scale into a pointwise bound on the claimed strain readout.

Proposition 2 is a warning about item (3). For the full subgrid port, the shell power is not controlled by $\|\nabla u\|_{L^\infty}$ times the shell energy at all, because the high–high source can feed an almost empty shell.

6.3 Consequence: localization is the lever

Theorem 5 identifies what a capacity route must avoid. If the admissible inputs at depth j can concentrate in one cell, a uniform certificate inherits the full strength of a pointwise bound. One way to keep a capacity certificate from inheriting that strength is to restrict the admissible inputs so that they cannot concentrate. This restriction is a proposal, not a consequence of the theorem: other restrictions of the input class could also block the basis tests. The next section makes the anti-localization restriction quantitative. Excluding localized tests only removes the obstruction. It does not by itself provide ICAP, WORK, DIV, or a continuation criterion.

7 Anti-localization and SBT-legal data

Section 6 showed that localized admissible inputs make a uniform capacity certificate as strong as a pointwise bound. This section states the restriction that removes those inputs, proves that it follows from a channel structure, and turns it into a certificate that can be computed on initial data.

The word "SBT-legal" is used in two senses. *Analytically*, it names a restricted problem in which the admissible interface inputs at each scale cannot concentrate. This is a hypothesis on admissible signals, not a theorem about Navier–Stokes. *Operationally*, it names a certificate that tests this restriction at a chosen resolution and time window. We use the certificate on initial data and as a diagnostic along short runs. We do not claim that passing it at $t = 0$ implies passing it later; that persistence question is open (section 9).

7.1 Anti-localization

Fix a shell j and let $B_j(x_0)$ be a ball of radius comparable to 2^{-j}. The *worst local energy fraction* of u in shell j is

$$L_j(u) := \sup_{x_0 \in \mathbb{T}^3} \frac{\int_{B_j(x_0)} |u_j(x)|^2 \, dx}{\|u_j\|_{L^2}^2} \in [0, 1], \tag{12}$$

with the convention $L_j(u) = 0$ when $u_j = 0$.

Definition 5 (Anti-localization at scale j)**.** A shell field $v \in \mathrm{Ran}(\Delta_j)$ is η_j*-anti-localized* if

$$\sup_{x_0 \in \mathbb{T}^3} \int_{B_j(x_0)} |v(x)|^2 \, dx \;\le\; \eta_j \, \|v\|_{L^2}^2. \tag{13}$$

A family of shell classes is *asymptotically anti-localized* if this holds with $\eta_j \to 0$.

The inequality needs no division, so it is meaningful for $v = 0$. If $\eta_j \to 0$, then at large j no admissible input can keep a fixed fraction $c_0 > 0$ of its energy in one ball $B_j(x_0)$: once $\eta_j < c_0$, the localized tests of section 6 are excluded. This blocks that particular argument. It provides no capacity bound, no work quantum and no divergence by itself.

7.2 Few spread-out channels force anti-localization

One way to obtain anti-localization is *channelization*: admissible shell inputs are combinations of a few channel shapes, each spread out in space. An interface that exposes only a few delocalized degrees of freedom cannot target a single cell. The following lemma is stated in a finite model, as it is mechanized.

Lemma 2 (Delocalized channels give anti-localization)**.** *Let ρ be a finite set of m channels, ι a finite set of grid cells, and $B \subset \iota$ one cell group. Let the channel shapes $\psi_r : \iota \to \mathbb{R}$ satisfy*

$$\sum_{i \in B} \psi_r(i)^2 \;\le\; \kappa\, \mu(B) \qquad \text{for every } r \in \rho,$$

with $\kappa \ge 0$ and $\mu(B) \ge 0$. Then every synthesized field $u = \sum_r a_r \psi_r$ satisfies

$$\sum_{i \in B} u(i)^2 \;\le\; m\, \kappa\, \mu(B) \sum_{r \in \rho} a_r^2. \tag{14}$$

If the channels are orthonormal, $\sum_i \psi_r(i)\psi_s(i) = \delta_{rs}$, then $\sum_{i \in \iota} u(i)^2 = \sum_r a_r^2$, and therefore $\sum_{i \in B} u(i)^2 \le m\, \kappa\, \mu(B) \sum_{i \in \iota} u(i)^2$.

Proof. Pointwise, $u(i)^2 \le m \sum_r a_r^2 \psi_r(i)^2$ by Cauchy–Schwarz. Summing over B and using the channel bound gives (14). For orthonormal channels, expanding the square and using the Gram identity gives Parseval. □

The bound holds for the cell group B named in the hypothesis. A bound uniform over all cells, as in (13), needs the channel hypothesis for every cell. Lean proves the coefficient form (`mass_cell_le_card_mul_deloc_mul_coeffEnergy`), the Parseval identity from the Gram matrix (`synthesis_energy_eq_coeffEnergy_of_orthonormal`) and the field-energy form (`mass_cell_le_card_mul_deloc_mul_fieldEnergy_of_orthonormal`), all in `formal/Fluids/AntiLocalization.lean`. The continuum version, with L^2 integrals over balls, follows from the same Cauchy–Schwarz argument and gives $\eta_j \le m_j \kappa\, \mu(B_j)$.

Scaling. A ball of radius $c\, 2^{-j}$ occupies a volume fraction $\mu(B_j) \sim 2^{-3j}$ of $\mathbb{T}^3$. With κ fixed, the returned constant $\eta_j \le m_j \kappa\, 2^{-3j}$ tends to zero exactly when $m_j 2^{-3j} \to 0$, for example when $m_j \lesssim j + 1$. A finite channel count at each depth is not enough without this growth condition.

7.3 Measured channels and their residual

In practice the channels are estimated from data, for example as the leading singular vectors of a matrix of shell snapshots. A measured span need not contain the data exactly, and the part left out can itself be localized. The following elementary bound charges it.

Lemma 3 (Residual-charged anti-localization)**.** *Let $\eta \ge 0$ and let $u = v + r \ne 0$, where v lies in a channel span whose fields satisfy (13) with constant η, and $r \perp v$ in L^2. Write $\delta = \|r\|_{L^2}^2 / \|u\|_{L^2}^2$. Then for every ball $B = B_j(x_0)$ of the family in (13)*

$$\frac{\int_B |u|^2}{\|u\|_{L^2}^2} \;\le\; \min\Big(1,\; \big(\sqrt{\min(\eta, 1)\,(1 - \delta)} + \sqrt{\delta}\big)^2\Big).$$

Proof. By the triangle inequality in $L^2(B)$, $\|u\|_{L^2(B)} \le \|v\|_{L^2(B)} + \|r\|_{L^2(B)} \le \sqrt{\min(\eta, 1)}\,\|v\|_{L^2} + \|r\|_{L^2}$, and orthogonality gives $\|v\|_{L^2}^2 = (1-\delta)\|u\|_{L^2}^2$. □

The returned bound tends to zero when both $\eta_j \to 0$ for the span and $\delta_j \to 0$ for the residual. These are sufficient conditions for this bound, not necessary ones for the actual fields: a residual can carry a fixed fraction of the energy and still be spread out. A rank that retains a fixed 95% of pooled snapshot energy controls neither quantity. It bounds the residual of the pooled data, not of each snapshot, and a small residual can be concentrated.

7.4 An operational certificate

We now define the computable notion used in the experiments. It is an admissibility filter for a restricted research question, not a criterion for the Clay problem: *do SBT-legal data evolve without Zeno cascades?*

Definition 6 (SBT-legality certificate)**.** On an N^3 grid, for each shell j let L_j be the largest fraction of the shell-j energy found in a periodic grid cube of side $w_j = \mathrm{round}(N/2^j)$ cells (zero for an empty shell), and let $f_j = (w_j/N)^3$ be the volume fraction of that cube. The *concentration factor* is

$$\kappa_j := L_j / f_j,$$

so a perfectly uniform shell has $\kappa_j = 1$ and a shell concentrated in one cube has $\kappa_j = 1/f_j$. Over a set of snapshots the certificate records $\max_t \kappa_j$, a *burst ratio* (the largest snapshot value of L_j divided by its mean over snapshots, zero for a shell that is empty in every snapshot) and, optionally, the number m_{95} of singular vectors needed to retain 95% of the pooled, uncentered snapshot energy. A field is *SBT-legal at thresholds* $(\kappa_\star, \mathrm{burst}_\star, m_\star)$ if every recorded statistic is valid and at most its threshold.

The certificate uses grid cubes, whereas (13) uses balls; comparing their constants requires a geometric adapter. The certificate tool (`ns_spt_legal_certify.py`; file names use the abbreviation "spt") writes a machine-readable JSON record with its configuration. It rejects invalid input instead of passing it: nonfinite or non-real spectra, inconsistent shell counts, factors that are not the stated ratio of raw fraction to baseline, and missing or non-integer ranks when ranks are required.

7.5 The certificate is not vacuous

A legal baseline and an illegal construction make the point. The baseline is a smooth random band-limited field of the kind used throughout the demos. The illegal field is a divergence-free blob placed in a single dyadic shell ($j = 3$) and concentrated near one point (`illegal_initial_conditions.py`), rescaled to the energy of the baseline. Band-limiting alone does not prevent concentration: the illegal field is band-limited by construction.

Remark 3 (What this does not claim)**.** We do not prove that Navier–Stokes preserves SBT-legality, nor that every smooth initial datum is legal. Anti-localization is one proposed way to avoid the obstruction of theorem 5; the theorem does not show that it is the only one. Here we state that hypothesis precisely, mechanize its finite inequality logic, and provide a reproducible certificate that tests it on concrete data.

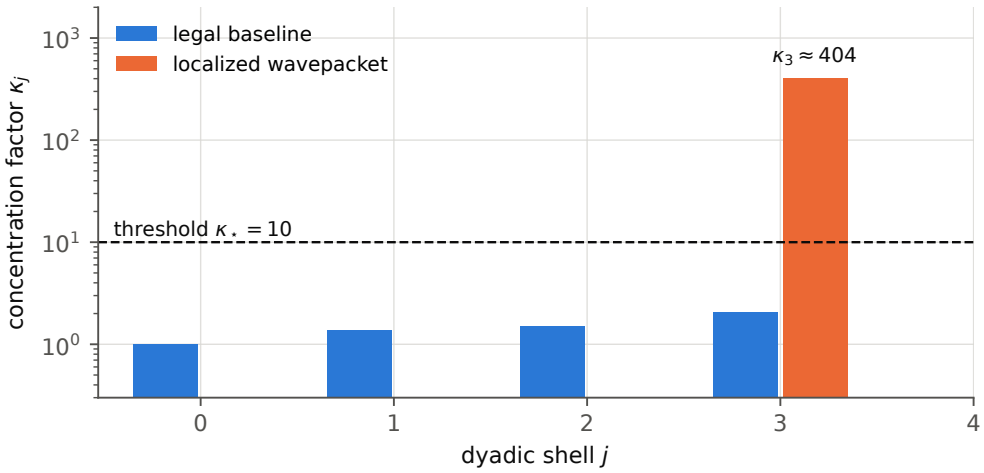


Figure 4: **Legal and illegal initial data under the same certificate** ($N = 32$, $t = 0$). The legal baseline has κ_j between 1.0 and 2.1 in every occupied shell. The localized wavepacket in shell $j = 3$ reaches $\kappa_3 \approx 404$, forty times the threshold $\kappa_\star = 10$. The shell $j = 4$ is empty in both fields, because the baseline is band-limited to $|k| \leq 10$ and the illegal field occupies only $j = 3$; empty shells are reported as zero. Note the logarithmic scale.

8 Diagnostics

The experiments in this section are diagnostics, not proofs. They check that the tooling computes the quantities of the scaffold consistently, and they ask whether the anti-localization restriction is plausible in tested regimes. Every artifact is produced by a script listed in the manifest `paper/figures_manifest.json` (appendix E). Each script prints its configuration and writes the plotted arrays to a companion `.npz` file, so every figure can be re-plotted from data, and the fast producers are checked for byte-identical output on rerun in a fixed software environment. Commands are in appendix D; the solver is described in appendix C.

The channel diagnostics and the sweep use the 2π-periodic box, a random band-limited initial field ($k_{\text{init}} = 10$), low-wavenumber forcing ($k_{\text{force}} = 2$, amplitude 10^{-2}), $\alpha = 2$, $\mu = 0$, time step 5×10^{-4} and final time 0.6, with the first 40% of each run discarded as burn-in in the channel diagnostics.

8.1 Channel count and delocalization

The finite-channel route to anti-localization (lemma 2) needs two things: few channels, and channels that are spread out. For shells $j = 1, \ldots, 4$ on a 32^3 grid with $\nu = 10^{-3}$ and three seeds, we collect shell snapshots and compute the rank m_{95} needed to retain 95% of their uncentered squared norm, both within sliding windows of 20 snapshots and after pooling all snapshots across time and seeds (figure 5a). Every window is effectively one-dimensional ($m_{95} = 1$ in all 15 windows per shell), while the pooled rank is 3 in every shell. A pooled rank above the windowed one is what one expects when a dominant channel rotates over time: each window sees one direction, and the pooled data span a few.

For the pooled channels we measure, in each shell, the largest local energy fraction of the channel shapes and the resulting span bound $\hat{\eta}_j$ of lemma 2 (figure 5b, solid). It falls from 0.56 at $j = 1$ to 0.0024 at $j = 4$. The span is not the data, however. The worst snapshot leaves a fraction 0.19 of

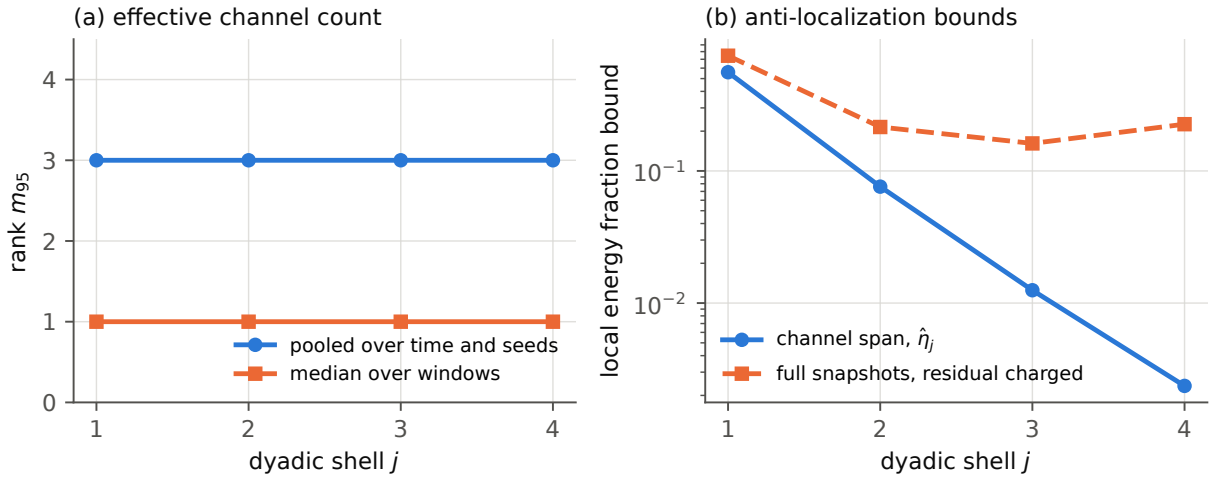


Figure 5: **Channel structure of the shells** (32^3, $\nu = 10^{-3}$, three seeds). (a) Rank m_{95} within windows of 20 snapshots (median and interquartile band, all equal to one) and after pooling. (b) Anti-localization bound for the retained channel span ($\hat{\eta}_j$) and for full snapshots once the discarded residual is charged (lemma 3). The gap at $j = 4$ is due to the residual.

its energy outside the retained span at $j = 4$, and charging that residual through lemma 3 gives a full-snapshot bound of 0.23 (dashed). At the finest shell the residual, not the span, controls the bound. A fixed 95% retention therefore does not show asymptotic anti-localization. For this residual-charged bound to establish it, both $\hat{\eta}_j \to 0$ and a residual fraction tending to zero would be needed.

8.2 Anti-localization sweep

To see whether shells avoid localization along the flow in the tested regimes, we run the certificate of definition 6 on 18 short forced runs: resolutions $N \in \{32, 48\}$, viscosities $\nu \in \{0.004, 0.002, 0.001\}$ and three seeds, recording snapshots every 30 steps. All 18 runs complete, so coverage is complete. Figure 6 shows the worst values over the sweep. The largest concentration factor is $\max_j \kappa_j \approx 5.79$ (at $j = 4$) and the largest burst ratio is 1.17, well below the thresholds $\kappa_\star = 10$ and $\text{burst}_\star = 50$; the largest rank m_{95} in any run is 2. Shell $j = 5$ exists only on the 48^3 grid and is empty after de-aliasing, so it is reported as zero; this expresses the empty-shell convention, not a high-frequency result.

A related sweep in the repository (`nsbu29_antilocalization_sweep.py`) reports a channel-normalized factor $\kappa_j/(j+1)$ instead of the raw factor; on the same runs its maximum is 1.16. The two factors answer different questions and should not be compared to the same threshold.

8.3 Route-capacity and frequency-response diagnostics

The route-capacity diagnostics (section 4) extract capacity-like ratios along a ladder of truncations ($N = 8, 12, 16, 24, 32$; $k_{\text{init}} = 3$, $\nu = 10^{-3}$, final time 0.2) and compare a direct elimination with an explicit two-stage nested composition. For the instantaneous field-level composition the two agree to round-off: the measured mismatch is of order 10^{-20}, as the exact nesting of sharp projections predicts. The sampled step factors of the capacity ratio lie between 0.60 and 0.71 over the three available steps; two step factors are unavailable, one because there is no preceding level and one

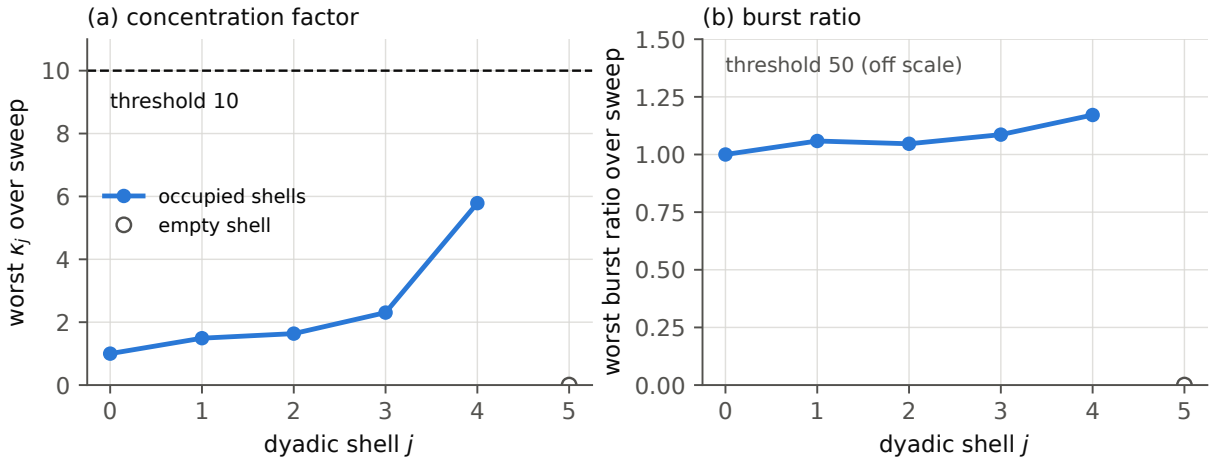


Figure 6: **Anti-localization sweep** (18 runs, complete coverage). Worst concentration factor κ_j and worst burst ratio in each shell over all runs and snapshots. Values stay far below the thresholds $\kappa_\star = 10$ and $\text{burst}_\star = 50$ in the tested window. This supports the plausibility of an anti-localization restriction in these regimes; it is not a worst-case statement about Navier–Stokes.

because the preceding capacity is zero. These are finite-window samples. They do not establish bounded mismatch for the packaged history map or a polynomial law.

The optional frequency-response diagnostics (32^3 run with $k_{\text{init}} = 8$, $\nu = 10^{-3}$, final time 0.6, coarse cutoff at 16^3) fit a linear time-invariant response to the subgrid term. They use the *extracted*-power convention, opposite in sign to (2). In that convention the real part of the measured response is mostly negative in two of the three tested shells, so the response is not passive there, and only the remaining shell is eligible for a passive rational fit. Its root-mean-square error is about 0.24; with each frequency's residual weighted by $\sqrt{d_i / \max_i d_i}$, where d_i is the shell input power at that frequency (the denominator of the estimate), it is about 1.8×10^{-3}. The fit misses much of the high-frequency data. Nonnegative fitted coefficients certify the passivity of the chosen rational model, not of the nonlinear Navier–Stokes bridge.

8.4 Legal and illegal data

Figure 4 in section 7 shows that the certificate is not vacuous: a band-limited but localized shell field fails it by a factor of forty, while the random baseline passes.

9 Discussion and open problems

What the scaffold establishes. The inequality logic of a No-Zeno argument is mechanized (section 3), together with two abstract results that bear on the capacity: route stability limits capacity growth to a polynomial (section 4), and, in a finite model, a uniform capacity certificate over localization-rich inputs is equivalent to a pointwise gain bound (section 6). On the Navier–Stokes side, the packaged bridge has a precise definition with an exact return on strong solutions (proposition 1). Energy alone gives a valid but exponentially growing capacity, and an explicit field rules out a shell-only strain bound for the full subgrid port (section 5). The anti-localization hypothesis that removes the finite obstruction follows from few spread-out channels, and a residual-charged version

applies to measured channels (section 7). A manifest-driven pipeline reproduces every numerical figure.

What it does not claim. We do not prove global regularity for the three-dimensional Navier–Stokes equations. None of the Navier–Stokes-facing hypotheses below is discharged by the Lean proofs, by the analytic lemmas or by the certificates. The emphasis is on making the chain explicit, and on showing where a capacity route needs a genuinely new inequality.

Why capacity is not a free lunch. A uniform capacity certificate bounds the positive work over all admissible inputs. In the finite model, once those inputs include localized tests, the certificate and a pointwise gain bound are the same statement. The always-available energy bound gives capacity growth $2^{5j/2}$, whose reciprocals have a finite sum. A capacity route therefore has to look beyond energy, for instance to a stronger source of capacity, to a restriction of the admissible inputs, or to both.

The restricted problem. The scaffold suggests studying SBT-legal data, for which the admissible inputs are anti-localized at every scale. Anti-localization removes the localized tests. It does not by itself provide a capacity bound, a work quantum, a divergence or a continuation criterion.

9.1 Open hypotheses

Each of the following is open here. A regularity proof along these lines would need items 1, 2 and 5. Items 3 and 4 concern the proposed route to item 2 and the obstruction it is meant to avoid.

1. **Work quantum for an actual frontier.** A storage functional for the eliminated scales that satisfies the passivity inequality along Navier–Stokes solutions, with attained crossings, so that lemma 1 applies.

2. **A capacity with divergent reciprocals.** An ICAP for the actual bridge with $\Lambda(j)$ growing at most linearly, for instance from a channel decomposition with $m_j \lesssim j+1$ atoms and a uniform per-atom constant, or from route stability of the actual packaging maps.

3. **Anti-localization as a theorem.** A proof that the admissible inputs of the actual bridge satisfy (13) with $\eta_j \to 0$, or a counterexample showing that intermittency defeats it, together with persistence of legality under the flow.

4. **The continuum no-go passage.** The three adapters listed in section 6, if one wants the finite obstruction to hold for the actual bridge.

5. **Return to the PDE (HL-NS-ZENO).** A proof that blow-up at a finite time T^* would force the chosen frontier to cross every level before T^*. At each fixed smooth time the high Sobolev tail $\sum_{\ell \geq J} 2^{2k\ell}\|u_\ell\|_{L^2}^2$ tends to zero as $J \to \infty$; the connection between the loss of a uniform bound near T^* and the storage frontier is not proved.

A related, also unproved, route would show that blow-up forces a quantified failure of passivity or a growing route mismatch. Finite-resolution diagnostics can look for such failures but do not prove them.

9.2 Further work

Theory.

- Prove or refute a channelization or delocalization statement for the actual bridge that excludes wavepacket-like admissible inputs.
- Determine whether such an exclusion can be derived from the structure of Navier–Stokes without assuming a classical peak bound [2, 10].
- Look for capacity sources beyond energy, including cumulative bounds over time windows, to which the counterexample of proposition 2 does not apply.
- Mechanize further interface-calculus steps as they enter the scaffold, including alternative discrete feasibility models.

Numerics.

- Extend the diagnostics to higher resolution, longer horizons and broader forcing and viscosity ranges, and quantify the robustness of the channel and anti-localization statistics.
- Stress-test the certificate with multi-shell, multi-site and intermittent constructions.
- Measure the sensitivity of κ_j, the residual fraction and the ranks to estimator choices such as window size, pooling and basis.

A Mechanized theorem index

All Lean statements are discrete or abstract: sequences indexed by $\mathbb{N}$, finite sums, finite index sets, real inner-product spaces and real functions of one variable. None of them formalizes a continuum PDE, a Sobolev estimate, a storage frontier or a limit $\mu \to 0$. The library builds with Lean 4 and mathlib v4.8.0 (`lake build` in `formal/`). The file `formal/ScaffoldAxiomAudit.lean` prints the transitive axioms of the 31 declarations below; every one depends only on `propext`, `Classical.choice` and `Quot.sound`. There is no `sorry` and no added axiom. All names live in the namespace `Fluids`.

Lean declaration	**Statement (paper reference)**
`Zeno.lean`	
`le_t0_add_sum_range_of_increments_ge`	$\Phi(n) \le t(n{+}1) - t(n)$ for all n implies $t(n) \ge t(0) + \sum_{k<n} \Phi(k)$.
`monotone_of_increments_ge_of_nonneg`	With $\Phi \ge 0$, the crossing times are nondecreasing.
`no_finite_upper_bound_of_increments_ge_of_sum_unbounded`	Theorem 1.
`no_finite_upper_bound_dyadic`	If $\Phi(2^j) \le t(j{+}1) - t(j)$ and the $\Phi(2^j)$ have unbounded partial sums, t has no finite upper bound.
`ZenoWorkCap.lean`	
`workcap_div_le_increment`	Theorem 2.
`increments_ge_of_workcap`	The same bound for every n.
`no_finite_upper_bound_of_workcap_of_sum_unbounded`	Theorem 3.
`no_finite_upper_bound_dyadic_of_workcap`	Dyadic version of theorem 3 (section 3).
`RouteCapacity.lean`	
`telescoping_pow`	One telescoping step, $((n+2)/(n+1))^p (n+1)^p = (n+2)^p$.
`route_capacity_poly_bound`	Theorem 4, with the tame factor substituted in the recurrence.
`route_capacity_poly_bound_of_bounded_sum`	The same, with bounded mismatch sums.
`route_capacity_poly_bound_of_tame_factor`	Theorem 4 from an actual factor $r(n)$ and $C \ge 0$.
`route_capacity_poly_bound_of_tame_factor_of_bounded_sum`	Its bounded-mismatch form, $C(n) \le (C(0) + E)(n+1)^p$.
`ECTCap.lean`	
`max_add_le_add_max`	$\max(x+y, 0) \le \max(x, 0) + \max(y, 0)$.
`max_sum_le_sum_max`	The same for finite sums (section 4).
`icap_sum_le_card_mul`	Per-atom bounds $w_i \le \Lambda_0 E$ sum to at most $\lvert s\rvert \Lambda_0 E$ (section 4).
`IcapNoGo.lean`	
`icap_implies_pointwise_pos_bound_of_basis_tests`	Theorem 5 with ICAP on all inputs.
`icap_implies_pointwise_pos_bound_on_feasible_basis_tests`	Theorem 5 with ICAP only on a feasible class containing the basis tests.
`AntiLocalization.lean`	
`mass_cell_le_card_mul_deloc_mul_coeffEnergy`	Lemma 2, coefficient form (14).
`synthesis_energy_eq_coeffEnergy_of_orthonormal`	Parseval from the Gram identity of orthonormal channels.
`mass_cell_le_card_mul_deloc_mul_fieldEnergy_of_orthonormal`	Lemma 2, field-energy form.

Lean declaration	Statement (paper reference)
`Energy.lean`	
`inner_rhs_self`	For $\mathrm{rhs}(u) = -B(u,u) - \nu Au - \mu A_\alpha u$ with $\langle B(u,u), u\rangle = 0$: $\langle \mathrm{rhs}(u), u\rangle = -\nu\langle Au, u\rangle - \mu\langle A_\alpha u, u\rangle$.
`inner_rhs_self_le_zero`	With positive semidefinite A, A_α and $\nu, \mu \geq 0$, this pairing is ≤ 0 (appendix B).
`Difference.lean`	
`inner_rhsDiff_self`	Pairing identity for $-\nu Aw - \mu A_\alpha w + g$.
`inner_rhsDiff_self_le_inner_g`	With positive semidefinite A, A_α and $\nu, \mu \geq 0$, that pairing is at most $\langle g, w\rangle$.
`hasDerivAt_normSq`	Derivative of $\|w(\tau)\|^2$ along a differentiable curve.
`deriv_normSq_le_two_inner_g`	Under the same hypotheses, if w is differentiable at t with $w'(t) = -\nu Aw(t) - \mu A_\alpha w(t) + g(t)$, the derivative value $2\langle w'(t), w(t)\rangle$ is at most $2\langle g(t), w(t)\rangle$.
`deriv_normSq_le_two_inner_g_at_rhs`	Under the same hypotheses, $\frac{d}{dt}\|w(t)\|^2 \leq 2\langle g(t), w(t)\rangle$ for the curve's norm square itself (appendix B).
`Multiplier.lean`	
`deriv_nonneg_of_nonneg_right`	$f(0) = 0$, $f'(0) = \nu$ and $f \geq 0$ on a right neighbourhood of 0 give $\nu \geq 0$.
`viscosity_coeff_nonneg`	The same, read as nonnegativity of a leading viscosity coefficient (appendix B).
`fixed_mode_passivity_counterexample`	$\ell(r) = r^2 - r/2$ has $\ell(0) = 0$, $\ell(r) \geq 0$ for $r \geq 1$ and $\ell'(0) = -1/2$ (appendix B).

B The regularized model behind the simulations

The simulations integrate the hyperviscous equation

$$\partial_t u + \mathcal{P}\big((u\cdot\nabla)u\big) + \nu\Lambda^2 u + \mu\Lambda^{2\alpha}u = \mathcal{P}f, \qquad \Lambda = (-\Delta)^{1/2}, \tag{15}$$

on a periodic box, where the gradient part of a force f is absorbed into the pressure, with $\mu = 0$ in most runs and $\mu > 0$ in the convergence check. This appendix records what is known about (15) at the scope used here. The proofs are analytic; the Lean statements in `Energy.lean`, `Difference.lean` and `Multiplier.lean` only encode the abstract algebra noted below.

Theorem 6 (Fixed-μ global regularity)**.** *Let $\nu \geq 0$, $\mu > 0$ and $\alpha \geq 5/4$. For smooth divergence-free initial data and smooth forcing, (15) has a unique global smooth solution.*

This is the classical hyperviscous result [6]. We use the critical energy method of Tao [11, Section 2], adapted to the torus.

Reduction to zero mean. The force enters only through $\mathcal{P}f$, whose spatial mean $\bar f(t)$ moves the mean velocity $m(t) = \overline{u_0} + \int_0^t \bar f$. With $X'(t) = m(t)$, $X(0) = 0$, the field $v(t,x) = u(t, x+X(t)) - m(t)$ solves the same equation with zero-mean data and the zero-mean, solenoidal force $\mathcal{P}f(t, x+X(t)) - \bar f(t)$. Translation preserves every Sobolev norm, and the mean is controlled separately: $\|u(t)\|_{H^m}^2 = \|v(t)\|_{H^m}^2 + |\mathbb{T}^3|\,|m(t)|^2$. All estimates below are for zero-mean v, which we again call u.

Local theory. The semigroup $e^{-t(\nu\Lambda^2 + \mu\Lambda^{2\alpha})}$ gains one derivative at the cost $C(1 + (\mu t)^{-1/(2\alpha)})$, which is integrable at $t = 0$ because $1/(2\alpha) < 1$. The Duhamel map is therefore a contraction in $C([0,\delta]; H^m)$, $m \geq 3$, so a uniform H^m bound before a putative maximal time allows a restart beyond it.

Energy. Nonlinear cancellation and Young's inequality give $\|u(t)\|_{L^2}^2 + \mu \int_0^t \|\Lambda^\alpha u\|_{L^2}^2 \le \|u_0\|_{L^2}^2 + \mu^{-1} \int_0^t \|\Lambda^{-\alpha} f\|_{L^2}^2$, so $a(t) = \|\Lambda^{5/4} u(t)\|_{L^2}$ is square-integrable on every finite interval.
Higher energy. After differentiating m times, the top transport term cancels. Every remaining term has the form $\int (\nabla^{j} u)(\nabla^{j_1} u)(\nabla^{j_2} u)$ with $j_1 + j_2 = j + 1$. The finite-exponent embeddings $\|\nabla u\|_{L^{12/5}} \lesssim \|\Lambda^{5/4} u\|_{L^2}$ and $\|\nabla^j u\|_{L^{12}} \lesssim \|\Lambda^{5/4} \nabla^j u\|_{L^2}$, Gagliardo–Nirenberg interpolation and Hölder bound these terms by $C_m\, a(t)\, \|u\|_{H^m}\, \|\Lambda^\alpha u\|_{H^m}$. Absorbing half of the dissipation gives $X' \le (1 + C_m \mu^{-1} a(t)^2) X + \|f\|_{H^m}^2$ for $X = \|u\|_{H^m}^2$. The coefficient is integrable, so Grönwall bounds X on every finite interval.
The constants degenerate as $\mu \to 0$. The theorem says nothing about $\mu = 0$.

Theorem 7 (Conditional recovery at rate $O(\mu)$)**.** *Fix $\nu > 0$, $\alpha \ge 5/4$ and common smooth data and forcing. Suppose the Navier–Stokes solution v ($\mu = 0$) is smooth on a given interval $[0, T]$, and let u_μ be the solution of (15). For every integer $m \ge 3$ there are K_m and $\mu_m > 0$, depending on v, T, the box, m, α and ν but not on μ, such that*

$$\sup_{0 \le t \le T} \|u_\mu(t) - v(t)\|_{H^m} \le K_m\, \mu \qquad \text{for } 0 < \mu \le \mu_m.$$

Sketch. The difference $w = u_\mu - v$ starts at zero and satisfies $\partial_t w + \nu \Lambda^2 w + \mu \Lambda^{2\alpha} w + B(v, w) + B(w, v) + B(w, w) = -\mu \Lambda^{2\alpha} v$, where $B(a, b) = \mathcal{P}((a \cdot \nabla) b)$. The regularization acts as dissipation on w plus a source that depends only on the smooth reference. Transport cancellation, product estimates and $H^m \subset W^{1,\infty}$ give $\frac{1}{2} X' \le C_m (M + \sqrt{X}) X + \mu F \sqrt{X}$ for $X = \|w\|_{H^m}^2$, $M = \sup_{[0,T]} \|v\|_{H^{m+1}}$ and $F = \|\Lambda^{2\alpha} v\|_{H^m}$. While $X \le 1$ this becomes $X' \le A_m X + \mu^2 F^2$, so $X(t) \le \mu^2 K_m^2$ with $K_m^2 = e^{A_m T} \int_0^T F^2$. For $\mu \le \mu_m = 1/(2 \max(1, K_m))$ this keeps $X \le 1/4$, so the region $X \le 1$ is never left. No bound on u_μ uniform in μ is assumed; the bootstrap produces it. □

The rate uses reference regularity $H^{m+2\alpha}$. Smoothness of v on $[0, T]$ is a premise. The theorem gives no continuation of Navier–Stokes beyond T. In Lean, `inner_rhs_self_le_zero` and the `Difference.lean` lemmas prove the abstract energy and difference-energy algebra in a real inner-product space. The Sobolev estimates and the bootstrap are not mechanized.

Numerical check. On a fixed 24^3 grid with $\nu = 10^{-3}$ and $\alpha = 2$, the relative low-mode difference between the $\mu > 0$ run and the $\mu = 0$ run, measured on the modes with $|k| \le 4$ at $t = 0.1$, decreases with μ (figure 7). This is a finite-dimensional consistency check on one grid. It does not supply the smooth reference of theorem 7.

A remark on low-frequency closure. The Six Birds packaging step suggests replacing the eliminated scales by an effective linear term. Suppose that term is a scalar, isotropic, time-local symbol $\ell(|k|^2)$ acting on solenoidal fields, with ℓ twice differentiable near zero, $\ell(0) = 0$ and $\ell(r) \ge 0$ for all small $r > 0$. Then $\ell'(0) \ge 0$ (`viscosity_coeff_nonneg`) and Taylor's formula gives $\ell(r) = \ell'(0)\, r + O(r^2)$. In an equation that already contains $\nu \Lambda^2$, the leading coefficient becomes $\nu + \ell'(0)$. Three caveats apply. The coefficient $\ell'(0)$ can vanish, as for $\ell(r) = r^2$. Positivity at the lattice frequencies of a fixed box does not imply positivity near zero: $\ell(r) = r^2 - r/2$ is positive at every $r = |k|^2 \ge 1$ but has $\ell'(0) = -1/2$, and Lean proves this counterexample (`fixed_mode_passivity_counterexample`). And an autonomous lift of a memory equation is local in the enlarged state, not in the velocity alone, so it does not supply the time-local premise. The actual bridge of definition 1 keeps its full history.

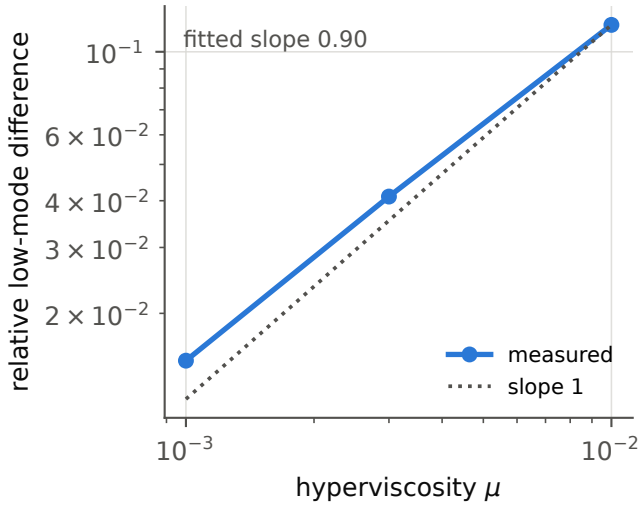


Figure 7: **Fixed-grid convergence in μ.** Relative low-mode difference between the hyperviscous run and the $\mu = 0$ run on the same 24^3 grid at $t = 0.1$, for three values of μ. The dotted line has slope one; the least-squares slope over the three points is 0.90.

C Numerical method

All three-dimensional experiments use a Fourier pseudo-spectral discretization [3] of (15) on $\mathbb{T}^3 = (\mathbb{R}/2\pi\mathbb{Z})^3$ with an $N \times N \times N$ grid. Fields are stored as Fourier coefficients $\hat{u}(k)$ and derivatives are computed spectrally. Incompressibility is enforced by the Leray projection $P(k) = I - k \otimes k/|k|^2$ for $k \neq 0$.

De-aliasing and real fields. Products are formed in physical space. The de-aliasing mask keeps the modes with $|k_x|, |k_y|, |k_z| < \frac{2}{3}\lfloor N/2 \rfloor$, with strict inequality [9]. A non-strict cutoff would let two retained modes alias onto a retained mode when N is divisible by three. Runs without de-aliasing still remove the ambiguous Nyquist faces, since an even grid identifies $+N/2$ with $-N/2$ and odd derivative symbols cannot be represented there. Restriction to a coarser grid also drops those faces, which preserves Hermitian symmetry, real fields and nested restriction. Spectra that are not those of real fields, and nonfinite states or forces, are rejected rather than silently repaired. With the strict cutoff the semi-discrete nonlinear term conserves energy; the test suite checks this cancellation.

Time integration. The linear part $L(k) = -(\nu|k|^2 + \mu|k|^{2\alpha})$ is integrated exactly. The classical fourth-order Runge–Kutta method is applied to the integrating-factor variable $e^{-Lt}\hat{u}$ (the Lawson scheme), with $E = e^{L\Delta t}$ and $E_{1/2} = e^{L\Delta t/2}$:

$$a = E_{1/2}\big(\hat{u} + \tfrac{\Delta t}{2} n_1\big), \quad b = E_{1/2}\hat{u} + \tfrac{\Delta t}{2} n_2, \quad c = E\hat{u} + \Delta t\, E_{1/2} n_3,$$
$$\hat{u}^{\,\mathrm{new}} = E\hat{u} + \tfrac{\Delta t}{6}\big(E n_1 + 2E_{1/2} n_2 + 2E_{1/2} n_3 + n_4\big),$$

where $n_1, \dots, n_4$ are the nonlinear and forcing terms at the stage states $\hat{u}, a, b, c$ and the corresponding stage times. Exact forced-linear and quadratic scalar solutions confirm fourth-order convergence in the tests. Runs end exactly at the requested time, taking a shorter final step when necessary, and record the terminal state.

Forcing. When forcing is on, a seeded random field is projected to divergence-free form, truncated to $|k| \le k_{\text{force}}$, de-aliased and rescaled to the configured root-mean-square amplitude. The forcing is deterministic given the seed and configuration.

Diagnostics. The local energy fractions are amplitude invariant and are computed after rescaling, so very small nonzero shells are not lost to underflow; an empty shell is assigned zero. Ranks m_{95} are counts of singular values of the uncentered snapshot matrix needed to reach 95% of its squared Frobenius norm. A sampled work-to-energy ratio is reported as a ratio over one sampled window, not as an extremum over admissible histories. Each producer records the configuration that applies to it (for evolution runs the grid, time step, final time, viscosities, exponent and forcing parameters, with separate coefficients for comparison runs; for certificates, the thresholds) in a printed `FIGURE_CONFIG` line and in its `.npz` file.

Scope. These simulations are diagnostics for the certificates and hinge quantities of the scaffold. Low-resolution runs are not evidence about continuum blow-up or regularity.

D Reproducing the results

All commands run from the repository root. The Python checks assume an environment with `python/requirements.txt` installed and `PYTHONPATH=python/src`.

Tests and proofs.

```
python -m pytest python/tests          # full suite, including slow tests
cd formal && lake build                 # Lean library
lake env lean ScaffoldAxiomAudit.lean   # transitive axioms of 31 declarations
```

Figures. The manifest `paper/figures_manifest.json` lists every producer. To regenerate all artifacts into `python/artifacts/paper/`, and then the main-text figures that are drawn from their `.npz` files:

```
python python/scripts/ns_paper_figures.py --include-slow --include-optional --clean
python python/scripts/ns_paper_publication_figures.py
```

The second command performs no simulation, but several of its figures use slow producers, so it needs the full set of artifacts from the first command. To re-run the fast producers and check that their `.npz` hashes are identical:

```
python python/scripts/ns_paper_figures.py --clean --verify-determinism
```

Determinism is checked in a fixed software environment. For reruns on other platforms we pin the thread count (`OMP_NUM_THREADS=1`) and recommend tolerance-based comparison.

Paper.

```
./check_paper.sh
```

E Figure manifest

Every numerical artifact is produced by a script listed in `paper/figures_manifest.json`. The listing below is generated from that manifest. Fast producers are checked for determinism; slow and optional producers are diagnostic enrichments, and none of them is needed for a mechanized claim. The main-text figures in `paper/figures/` are drawn from the `.npz` files listed here by `python/scripts/ns_paper_publication_figures.py`.

`fig_ns14_energy_ledger_3d` class: fast; **script:** `python/scripts/ns14_energy_ledger_3d.py`
png: `python/artifacts/paper/ns14_energy_ledger_3d.png`; **npz:** `python/artifacts/paper/ns14_energy_ledger_3d_data.npz`.

`fig_ns15_uv_completion_3d` class: fast; **script:** `python/scripts/ns15_uv_completion_turns_on_3d.py`
png: `python/artifacts/paper/ns15_uv_completion_3d.png`; **npz:** `python/artifacts/paper/ns15_uv_completion_3d_data.npz`.

`fig_ns16_mu_convergence_3d` class: fast; **script:** `python/scripts/ns16_mu_to_zero_convergence_3d.py`
png: `python/artifacts/paper/ns16_mu_convergence_3d.png`; **npz:** `python/artifacts/paper/ns16_mu_convergence_3d_data.npz`.

`fig_ns18_exponent_sanity` class: fast; **script:** `python/scripts/ns18_exponent_sanity_harness.py`
png: `python/artifacts/paper/ns18_exponent_sanity.png`; **npz:** `python/artifacts/paper/ns18_exponent_sanity_data.npz`.

`fig_nsbu12_toy_zeno_gallery` class: fast; **script:** `python/scripts/nsbu12_toy_zeno_gallery.py`
png: `python/artifacts/paper/nsbu12_toy_zeno_gallery.png`; **npz:** `python/artifacts/paper/nsbu12_toy_zeno_gallery_data.npz`.

`fig_ns_spt_illegal_vs_legal` class: fast; **script:** `python/scripts/ns_spt_illegal_ic_demo.py`
png: `python/artifacts/paper/ns_spt_illegal_vs_legal.png`; **npz:** `python/artifacts/paper/ns_spt_illegal_vs_legal_data.npz`.

`fig_ns_sgs_shell_counterexample` class: fast; **script:** `python/scripts/ns_sgs_shell_counterexample.py`
png: `python/artifacts/paper/ns_sgs_shell_counterexample.png`; **npz:** `python/artifacts/paper/ns_sgs_shell_counterexample_data.npz`.

`fig_nsbu21_route_capacity_numbers` class: slow; **script:** `python/scripts/nsbu21_route_capacity_numbers.py`
png: `python/artifacts/paper/nsbu21_route_capacity_numbers.png`; **npz:** `python/artifacts/paper/nsbu21_route_capacity_numbers_data.npz`.

`fig_nsbu23_route_capacity_hypothesis_dense` class: slow; **script:** `python/scripts/nsbu23_route_capacity_hypothesis_check.py`
png: `python/artifacts/paper/nsbu23_route_capacity_hypothesis_dense.png`; **npz:** `python/artifacts/paper/nsbu23_route_capacity_hypothesis_dense_data.npz`.

`fig_nsbu28_antilocalization` class: slow; **script:** `python/scripts/nsbu28_antilocalization_diagnostic.py`
png: `python/artifacts/paper/nsbu28_antilocalization.png`; **npz:** `python/artifacts/paper/nsbu28_antilocalization_data.npz`.

`fig_nsbu29_antilocalization_sweep` class: slow; **script:** `python/scripts/nsbu29_antilocalization_sweep.py`
png: `python/artifacts/paper/nsbu29_antilocalization_sweep.png`; **npz:** `python/artifacts/paper/nsbu29_antilocalization_sweep_data.npz`.

`fig_ns_chan00_channel_count` **class:** slow; **script:** `python/scripts/ns_chan00_channel_count.py`
png: `python/artifacts/paper/ns_channel_count.png`; **npz:** `python/artifacts/paper/ns_channel_count_data.npz`.

`fig_ns_chan01_channel_delocalization` **class:** slow; **script:** `python/scripts/ns_chan01_channel_delocalization.py`
png: `python/artifacts/paper/ns_channel_delocalization.png`; **npz:** `python/artifacts/paper/ns_channel_delocalization_data.npz`.

`fig_ns_spt_legal_sweep` **class:** slow; **script:** `python/scripts/ns_spt_legal_sweep.py`
png: `python/artifacts/paper/ns_spt_legal_sweep.png`; **npz:** `python/artifacts/paper/ns_spt_legal_sweep_data.npz`.

`fig_nsbu04_sgs_H` **class:** optional; **script:** `python/scripts/nsbu04_sgs_frequency_response.py`
png: `python/artifacts/paper/nsbu04_sgs_H.png`; **npz:** `python/artifacts/paper/nsbu04_sgs_H_data.npz`.

`fig_nsbu05_sgs_fit` **class:** optional; **script:** `python/scripts/nsbu05_sgs_realization_fit.py`
png: `python/artifacts/paper/nsbu05_sgs_fit.png`; **npz:** `python/artifacts/paper/nsbu05_sgs_fit_data.npz`.

F Version note

Version 1 of this preprint appeared on 4 March 2026. Version 2 (14 October 2026) follows a mathematical review of the supporting derivations, Lean library and numerical code. Its main changes are:

- the bridge is defined with sharp Fourier shells, with the sign of the supplied power fixed, and an exact return on strong solutions is proved (proposition 1);
- a section on what energy alone gives the capacity is added, including the counterexample to a shell-only strain bound (section 5);
- the no-go theorem is stated on a feasible class, and the continuum passage is listed as open;
- the divergence comparisons are stated with the correct quantifiers, and the work-quantum lemma states its regularity assumptions;
- the route-capacity theorem is connected to Lean through tame-factor adapters, and the anti-localization lemma through a Parseval identity; the theorem index lists all 31 audited declarations;
- the residual-charged anti-localization bound and the regularized-model appendix are added;
- the time stepping, de-aliasing, sign conventions and diagnostics of the numerical code are corrected, and all figures are regenerated; the illegal example's concentration factor is ≈ 404 in the corrected code.

The title and the main mechanized theorems are unchanged.

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